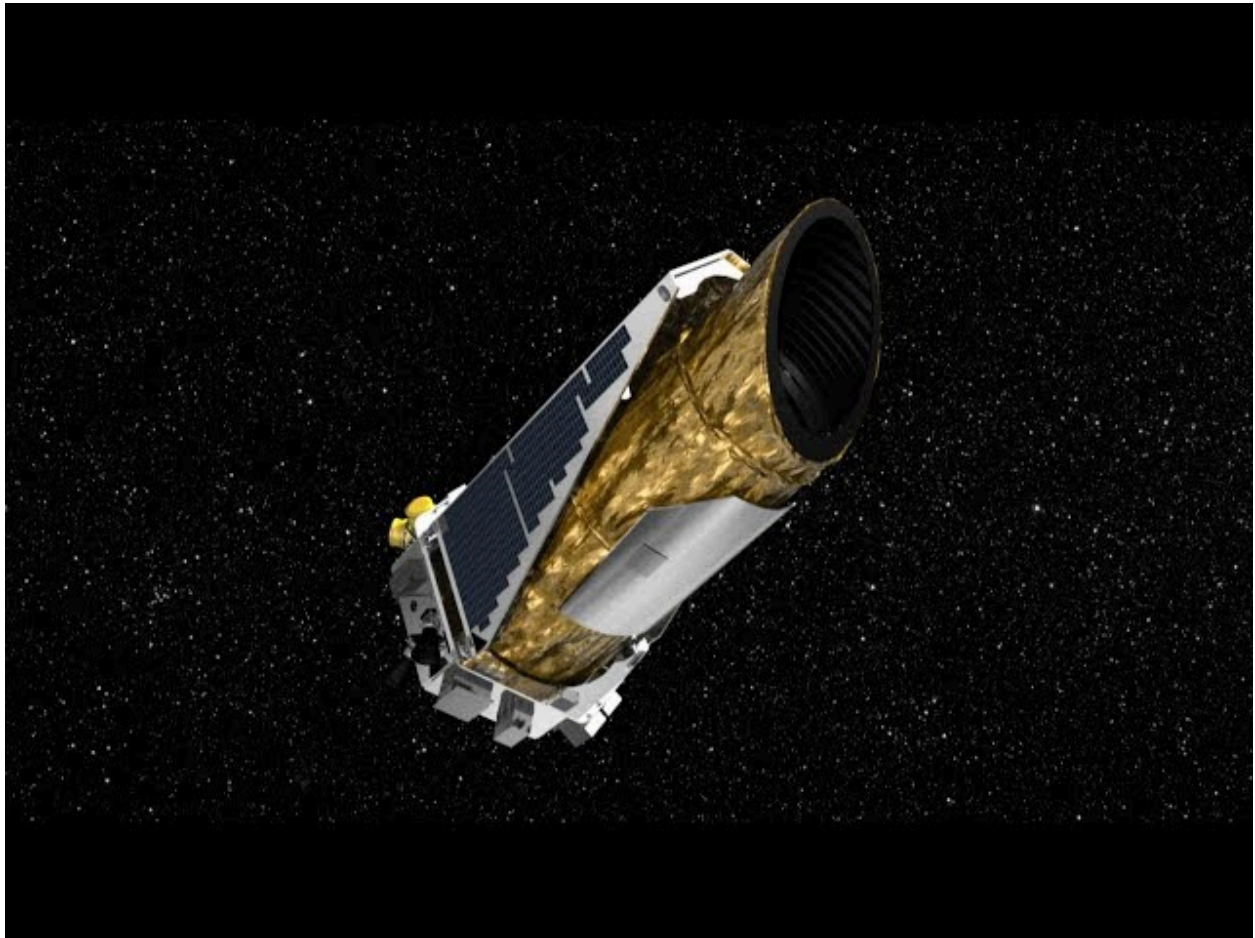


DS 100 - Final Project

By Aman, Naiem, Lucas



Background Information (AMAN)

NASA launched the Kepler Space Observatory in 2009. Its main purpose was to detect exoplanets. These exoplanets are planets that are outside our solar system. The Kepler Space Observatory determines possible exoplanets by measuring and tracking small drops in a stars' brightness of over 150,000 stars. A dip in the star's brightness means that there may be an object that might be a planet that is passing in front of its host star. This dip in the star's brightness is called a transit. Objects that have a transit kind of behavior are then labeled as Kepler Objects of Interest (KOI). By detecting and keeping a count of the number of transits, astronomers are able to identify potential exoplanets. But not every transit signal ends up being a real planet. Some transit signals come from binary stars, instrumental noise or some other astrophysical phenomena and therefore KOIs are then categorized as:

- Confirmed Planet: A scientifically justifiable planet
- Candidate: Might be a planet but is not confirmed/validated by scientists

- False positive: Originally thought to be a planet but with further investigation it is not

Over the time the Kepler Space Observatory has been operating, it has been able to make a large dataset of these KOIs along with specific features such as orbital period, transit depth and star temperature. Using this dataset scientists have been able continue exoplanet research even after the telescope shut down in 2018.

Dataset Overview (AMAN)

The dataset we use is NASA's Cumulative Kepler KOI table. This dataset has thousands of potential exoplanet candidates with more than 40 features for each one of the candidates. Some of these features are:

- Transit columns: depth, duration, impact parameter
- Orbital characteristics: period, semi-major axis
- Star properties: radius, temperature
- Detection data: model flags, SNR (signal to noise ratio)

The initial dataset had:

- 9,564 KOIs (rows)
- More than 50 numerical and categorical features

Some of the columns had non numerical values so while cleaning the data, we dropped those columns along with columns that had an excessive amount of missing data. After removing these columns, some of the remaining numeric features still had some missing values, so to be able to use these columns we filled in the missing values by using median imputation. Median imputation was very useful for us because it is robust to outliers and easy for us to understand. After filtering we were left with:

- Rows: 9564
- Columns: 47

and no missing values.

Defining the Problem (AMAN)

The research question we are focusing on in this project is:

Given the measurable properties of a Kepler Object of Interest (KOI), is it possible for us to predict whether it is likely to be a real exoplanet or a false positive?

As mentioned earlier, NASA provides 3 dispositions for KOIs: Confirmed, Candidate, False Positive. The candidate objects remain unverified until more research and investigation is done and since there isn't enough time and resources, scientists will have to prioritize which of the candidates are the most valuable candidates.

In order for us to approach this research question we converted the 3 dispositions into a binary classification task where:

- 1 = Real planet (Confirmed or Candidate)
- 0 = False positive

Using this binary classification, we can now apply and use machine learning models to learn patterns in the KOIs we currently have in our filtered dataset and then apply the patterns to the unconfirmed candidates. The binary classification can also help us assign confidence scores that will be able to help us identify high likelihood planets, high likelihood false positives, ambiguous objects (the ambiguous KOIs would require further studies for scientists).

Methodology (AMAN)

For our project we will be using the Random Forest classifier. This machine learning method is good for handling complex and noisy data which is perfect in the context of our dataset. Random forest is part of a group of algorithms that are known as ensemble methods. These algorithms do not rely on a single model but instead they combine the output of many individual learners. For random forests, the individual learners are called decision trees and they produce the final prediction. The design makes the model more robust as compared to an individual tree.

Random forest is a good model in our context because:

1. It is able to handle noisy data

Noisy data is any data that contains random or irrelevant values, errors, and inconsistencies that obscure the true patterns and relationships within a dataset. In our context, astronomical data is usually noisy because of several factors such as sensor limitations and environmental interference. If we use a single decision tree then it can become overly sensitive to this noisy data and may end up splitting on irrelevant or misleading patterns. By using random forest we are able to train each tree on a different random subset of the data and by randomly selecting a subset of columns/features at each split. Now because trees are seeing a slightly different version of the data, noise does not play a dominant role in the final prediction. If in any case a tree is influenced by the noise it will be outvoted by the rest of the ensemble.

2. Reduces overfitting

Overfitting is when a model memorizes the training data instead of actually learning the patterns in the data. This makes it so that it will perform perfectly with the data it has already seen but if we put it against unseen data, it will perform badly. Having a single tree will make it overly specific and it would start to memorize noise or misleading patterns in the training data instead of actually learning the patterns. Random forest is

able to fix this by training each tree on a different sample of data and also showing each tree a different subset of columns/features.

3. Is able to provide both predictions and probability estimates

Usually, simple classifiers like logistic regression only return a 0 or 1 label. Random forests are able to do that and also give predicted probabilities (eg. there is a 0.97 of being a planet) along with the distribution of the tree votes. This is much better than just returning a 0 or 1 label because in scientific applications the model is able to provide a confidence score which would be very useful in the context of our dataset.

4. Widely used and trusted scientifically

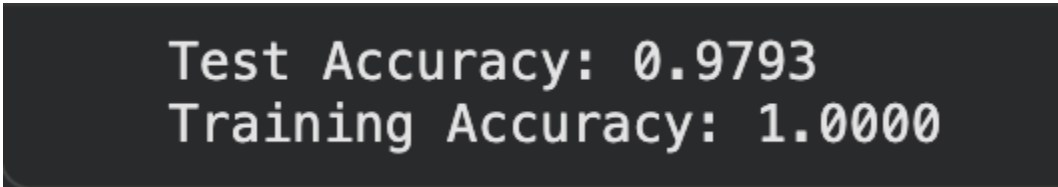
Random forest is known to be one of the most common machine learning algorithms that is used in scientific research. This is because it makes very few assumptions in the data and is grounded in well known statistics concepts like bootstrap aggregation, random subspace methods and ensemble averaging. This strong theoretical support behind the model gives many scientists confidence that the model outputs are backed by proven mathematically sound methods.

Analysis findings (Naïem and Aman)

After we cleaned the dataset, we shifted our focus to the analysis. Our analysis focuses on training, evaluating and then interpreting the random forest model using the cleaned dataset. We explored the model's predictive accuracy, the distribution of predictions for unconfirmed KOIs, the stability of these predictions across multiple runs, and the correlation between the columns/features and the likelihood of being a real exoplanet.

Model performance:

After splitting the dataset into 80% training data and 20% testing data we trained the random forest with 100 trees and got the following results:



Test Accuracy: 0.9793
Training Accuracy: 1.0000

The training accuracy tells us that the model will work and fit perfectly with the training data. The test accuracy was almost perfect and is able to generalize well and consistently reliably find out if an exoplanet is real or a false positive,

Prediction on candidate KOIs:

After training the data, we applied the model to 2248 candidate KOIs and got the following results:

Predicted as REAL PLANET: 1249 (55.6%)
Predicted as FALSE POSITIVE: 999 (44.4%)

And then got a list of the top 10 most likely planets and top 10 least likely planets:

```
Top 10 most likely real planets (highest confidence):
  kepoi_name  planet_probability  predicted_planet
153  K00841.03                1.0                1
210  K00892.02                1.0                1
303  K00935.04                1.0                1
483  K00141.01                1.0                1
574  K01085.01                1.0                1
613  K01030.01                1.0                1
645  K01165.01                1.0                1
671  K01239.02                1.0                1
845  K01276.01                1.0                1
846  K01276.02                1.0                1

Top 10 most likely false positives (lowest confidence):
  kepoi_name  planet_probability  predicted_planet
820  K01448.01                0.0                0
961  K01296.01                0.0                0
1241 K01539.01                0.0                0
1875 K02311.01                0.0                0
2874 K00347.01                0.0                0
5577 K01714.01                0.0                0
5746 K00364.01                0.0                0
6334 K05533.01                0.0                0
6434 K05807.01                0.0                0
6609 K05353.01                0.0                0
```

This ranked list of the most likely and the least likely planets makes it easier for scientists to know which KOIs they should prioritize their time and resources to. The candidate KOIs that have the highest confidence score should be prioritized while the candidate KOIs with the least confidence score are more likely to be noise or not a planet.

Measuring the prediction stability:

Due to random forest functioning by selecting a random subset of data and columns/features which can cause predictions to be different and vary between each time the code runs. We decided to analyze this variability instead of ignoring it. We trained 50 different random forest models where each of them have a different random seed.

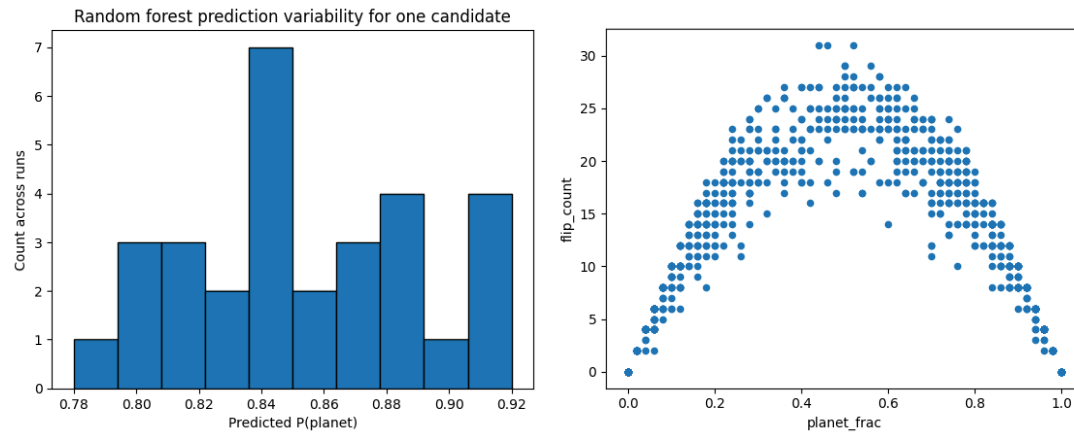
For each KOI we found out 'planet_frac'. This is the fraction of the 50 models that predicted "planet".

1.0 means that every run predicted planet

0.0 means that every run predicted false positive

around 0.5 means its vague/ambiguous

For each KOI we found 'flip_count'. This shows the number of times that the prediction changed in between each run and got the following results:



...	kepoi_name	planet_frac	flip_count
2256	K02626.01	0.44	31
1148	K01649.01	0.52	31
6328	K05406.01	0.46	31
3761	K04343.01	0.50	29
4631	K04810.01	0.56	29
1394	K02042.01	0.50	29
8070	K02564.02	0.50	28
2102	K02512.01	0.58	28
7703	K07072.01	0.52	28
7387	K04535.02	0.58	28
...			
...	kepoi_name	planet_frac	flip_count
2256	K02626.01	0.44	31
1148	K01649.01	0.52	31
6328	K05406.01	0.46	31
3761	K04343.01	0.50	29
4631	K04810.01	0.56	29
...	...	...	...
4275	K04587.01	0.78	20
6977	K06104.01	0.26	20
6554	K05842.01	0.22	20
7794	K06687.01	0.62	20
8581	K04668.01	0.26	20
220 rows x 3 columns			

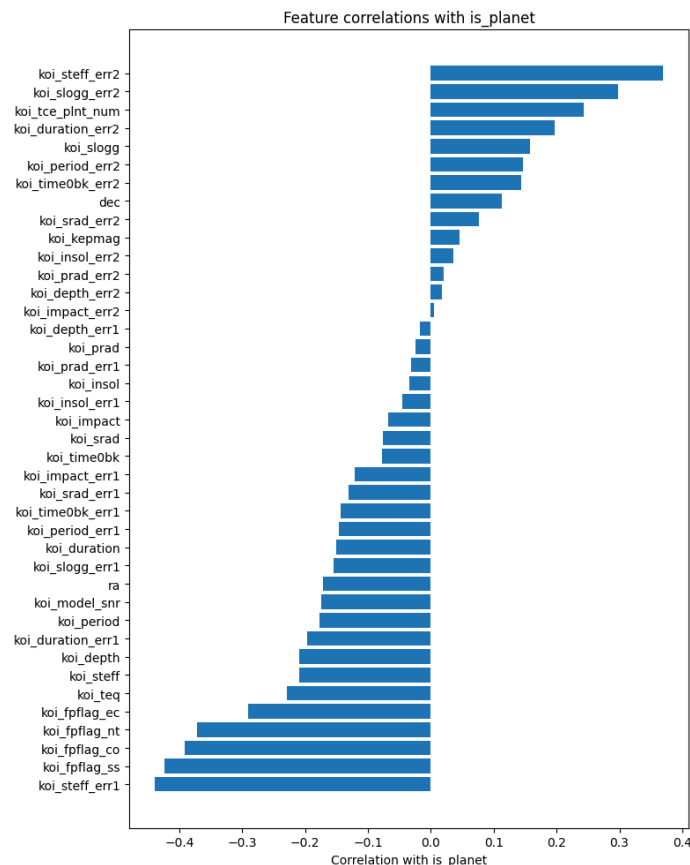
The results show that KOIs with a 'planet_frac' near 0 or 1 had 'flip_count' of 0 which means that those KOIs with that 'planet_frac' are highly stable. While the KOIs with a 'planet_frac' of around 0.5 had 'flip_count' of up to 31 which means that those KOIs with that 'planet_frac' are very uncertain. The overall shape forms a parabolic curve with a negative x^2 coefficient where the uncertainty is the highest in the middle. This curve helps researchers visualize that near predictions 0% or 100% can be trusted, while predictions ~50% should be followed-up on.

Feature Correlation:

After measuring the prediction stability we then started to calculate the correlation between every numeric feature and 'is_planet' and got the following results:

Our correlation analysis identified key features between real planets and false positives. Real exoplanets showed positive correlations with measurement uncertainties in stellar temperature (koi_staff_err2), surface gravity (koi_slogg_err2), and transit duration

(koi_duration_err2), suggesting that genuine planets are often found around fainter stars with imprecise measurements. Conversely, false positives were strongly associated with contamination flags (koi_fpflag_co, koi_fpflag_ec, koi_fpflag_ss) that indicate the signal originates from eclipsing binaries or nearby stars rather than true planetary transits



How Random Forest Helps Us Understand the Data (Naïem)

Random forest was very useful to us for understanding the data because:

- Reliability
 - We achieved a nearly perfect accuracy, at around ~98% accuracy on test data demonstrating strong prediction power of the Random Forest Model when it came to distinguishing real exoplanets and false positives.
 - With a perfect training accuracy and minimal overfitting, the model learned meaningful patterns
 - Even though the class imbalance was slightly significant, the model calculated fair predictions
- Identified ambiguous KOIs

- Revealed planetary and stellar characteristics most critical for exoplanet confirmation
- Help astronomers prioritize which candidates to focus on when funding stellar exploration
- Confidence scores
 - Probability scores, 0-1, provide confidence levels for each prediction, allowing for identification and prioritization of high confidence candidates for further observation
- Pattern Recognition
 - 43 different features simultaneously analyzed, detecting complex, non-linear relationships
 - 100 decision trees provided robust predictions
 - Messy data handling was feasible and was easily managed.

Random Forest Limitations (Naïem)

- Random Forests are hard to interpret; it is hard to trace the exact decision path for why a specific KOI was classified as an exoplanet or false positive. This can be extremely problematic when it comes to justifying our predictions to different audiences
- The computational expense of training hundreds of trees on large datasets is immense, requiring lots of memory and time. Prediction time scales with the increase in trees, making real time classification slower. Specifically, abnormally large datasets will take an absurdly long time to produce results without distributing computation.
- Highly correlated features affect Random Forest predictions a lot. This means that correlated features can artificially inflate scores for predictions, making it harder to identify other critical features. Highly correlated features tend to be selected for decision trees potentially significantly impacting prediction outcomes.

Further Improvements (Naïem)

- Change up the random forest hyperparameters
By trying a different number of trees, tree depths, and feature sampling strategies it might have been possible to improve the accuracy even more and also reduce the variance between runs.
- Use other advanced alternative models
When we were first researching this dataset we also found many other models in the process such as Gradient Boosting, XGBoost, and LightGBM. These models are sometimes able to outperform the random forest in some data sets so it would have been interesting to see how they would work on our dataset.
- Create other new columns/features/variables
Creating new features/variables by using ratios of other different features will help us find more insights from the data like patterns we were not able to see before and also increase the scope of our research question.

- Comparing to other simple baseline models
By comparing the random forest to other simpler models like the logistic regression or KNN would be able to show us the actual difference between these simpler models and the random forest and if the random forest is more useful than these other models.
- Reduce the redundancy
Some of the columns/features in the dataset were highly correlated and had very similar measurements. We can simplify the model more by removing or maybe even combining some of these features which can then potentially improve generalization.

Conclusion (Naïem)

This project displayed that Random Forest classification can effectively classify real exoplanets from false positives in the Kepler dataset accurately. Our model successfully classified 2,248 confirmed candidates, identifying 1,202 as likely real exoplanets and providing confidence scores that allow astronomers to plan follow-up observations on varying candidates showing the most uncertainty

Beyond prediction accuracy, our analysis provided specific scientific insights such as the correlation of measurement uncertainties in stellar properties to real planet discovery, while also highlighting that contamination flags can be reliable for the identification of false positives. This knowledge helps astronomers with future exoplanet research as they have effective detection strategies

Despite strong performances with data sets containing a heavy amount of features, limitations do exist within the Random Forest Model. With reduced interpretability, combined labels creating potential noise, and large potential for data leakage swaying results, the Random Forest model does have some faults when it comes to detailed analysis.

Ultimately, Random Forest proved to be an effective algorithm for classification, aggregating predictions from multiple decision trees to reduce overfitting, and handling noisy astronomical data. Its ability to provide binary classification scores coupled with feature importance rankings makes it particularly well-suited for scientific applications. While more complex deep learning models exist, Random Forest is able to balance performance and computational efficiency, demonstrating why it is a foundational tool in machine learning.

Member Contributions:

Aman: Background information, Dataset overview, Defining the problem, Methodology, Analysis findings, Jupyter notebook, created presentation slides.

Naiem: Analysis findings, How random forest helps understand data, Random forest limitations, Further improvements, Conclusion, Jupyter notebook, created presentation slides.

Lucas: Created presentation slides.